

Background and Problem Statement

- Deep learning classifiers for **medical imaging** are routinely deployed with a single forward pass, but these predictions are frequently **overconfident** and **poorly calibrated**.
- Calibration** is a safety-critical requirement in clinical medicine to prevent systems from creating a dangerous illusion of certainty.
- Poor calibration is compounded by modern cross-entropy training objectives, distribution shifts between scanners, and class-imbalanced medical datasets.
- Test-Time Augmentation (TTA)** improves robustness, but standard TTA assumes uniform weights, which is fundamentally flawed for medical imaging.



88.1% → 56.5%

PathMNIST accuracy collapse with 50 equal-weight TTA views (Medeiros D. N. (2026) - prior work)

a 31.6 pp degradation

Research Gaps

Accuracy Degradation

Naive equal-weight TTA frequently degrades accuracy because natural-image transformations can be clinically meaningless or impossible

Calibration Impact

The impact of TTA on predictive calibration has not been systematically studied.

Uncertainty Weighting

It is unknown whether uncertainty-based weighting can recover reliable confidence estimates across diverse modalities.

Mechanism Clarity

The specific conditions under which uncertainty-based weighting actually helps or hurts remain poorly understood

Key Contributions

Systematic Evaluation

Evaluated nine calibration-focused TTA fusion strategies across six distinct medical imaging modalities and three backbones.

Two-factor Explanation

Proposed a two-factor explanation demonstrating that calibration gains depend on entropy informativeness and augmentation safety.

Visualizing Confidence

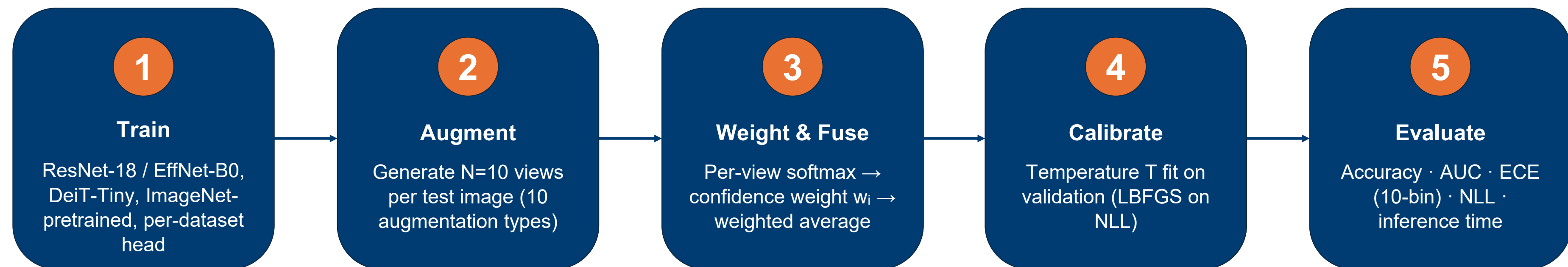
Introduced "Augmentation Confidence Strips" to expose which augmented views a model trusts or rejects

Practitioner Guide

Distilled empirical findings into an actionable selection guide for applying TTA under calibration constraints.

Methodology & System Pipeline

One trained model → N augmented views → confidence-weighted fusion → calibrated prediction

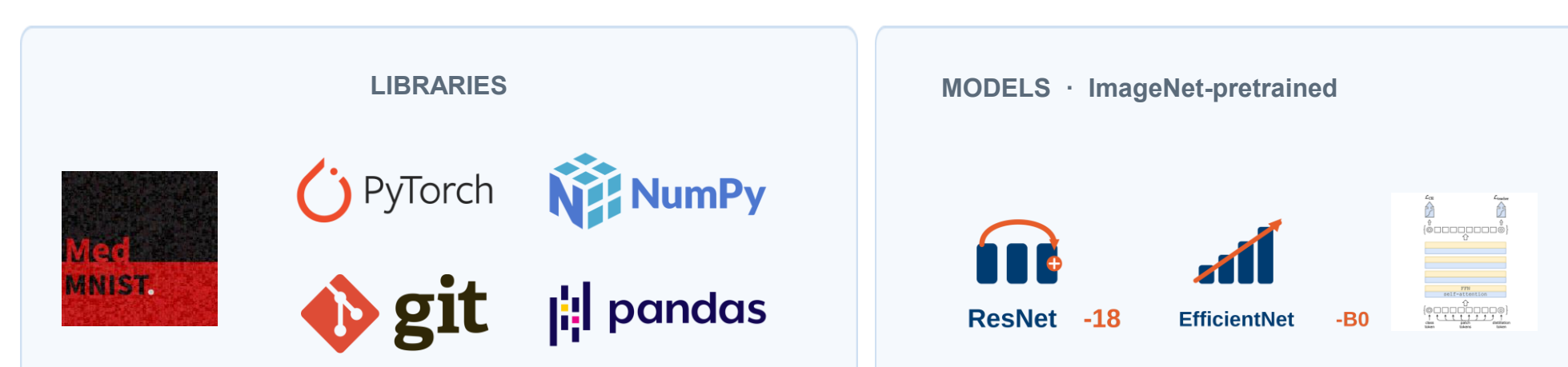


Used Datasets

Dataset	Modality	Task	Classes	Train size
PathMNIST	Colon pathology (histology)	Multi-class	9	~90,000
DermaMNIST	Dermatoscope (skin)	Multi-class	7	~7,000
PneumoniaMNIST	Chest X-ray	Binary	2	~4,700
BreastMNIST	Breast ultrasound	Binary	2	546
BloodMNIST	Blood-cell microscopy	Multi-class	8	~12,000
OrganAMNIST	Abdominal CT	Multi-class	11	~35,000

All from MedMNIST v2 · 64×64 · pip-installable & pre-split · spanning 5 imaging modalities, binary + multi-class tasks, and a wide size range (546 → ~90k) for a diverse benchmark.

Teck Stack



Fusion Strategies

Eight soft-weighting strategies below; a Top-K entropy-filter family (top3 / top5 / top7) is also reported as an ablation

Baseline TTA Every view weighted equally $w_i = \frac{1}{N}$	Variance-inv Literal proposal formula → negative finding $w_i = \frac{1}{var + \epsilon}$
Max-Prob Confident views dominate $w_i = \max(p_i)$	MC-Dropout Epistemic proxy, no augmentation <i>T stochastic passes</i>
Entropy Low-entropy views weigh more usual winner $w_i = e^{-H(p_i)}$	TS-only Temperature scaling, single pass which is nearly free <i>softmax(z / T)</i>
Variance Sharp views up (confidence-aligned) $w_i = \text{var}(p_i)$	TS + Entropy Best, most consistent calibration <i>entropy on z / T</i>

Results

Calibration Improved, Accuracy Maintained

+ 0.04 p

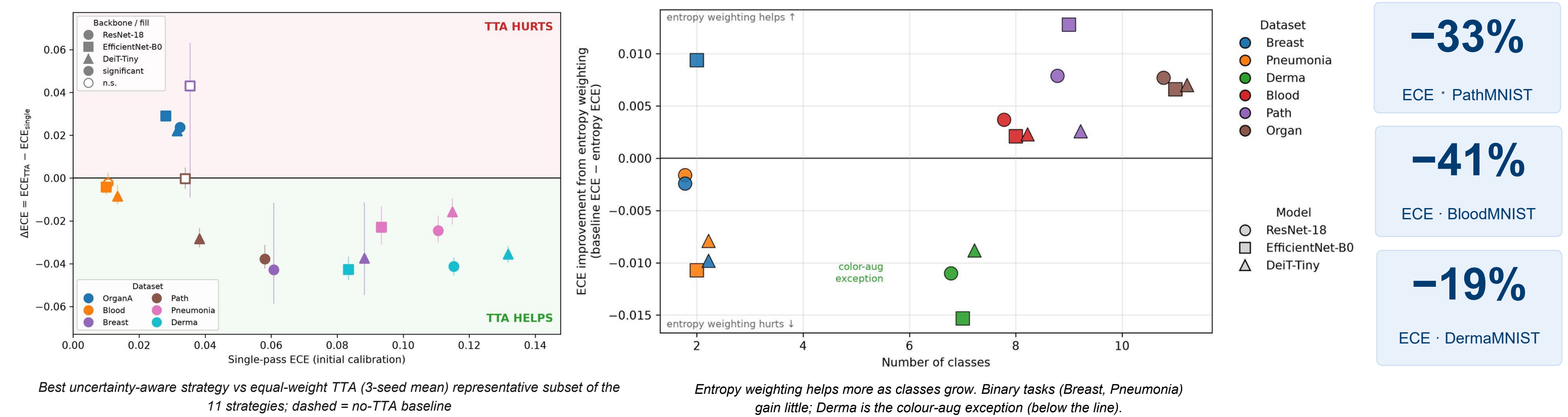
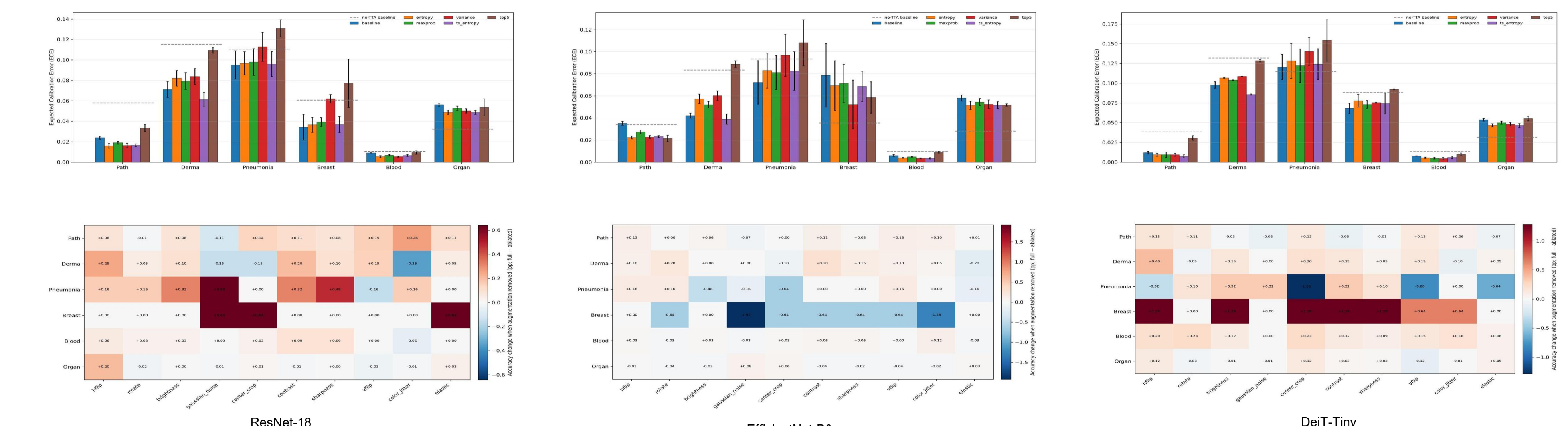
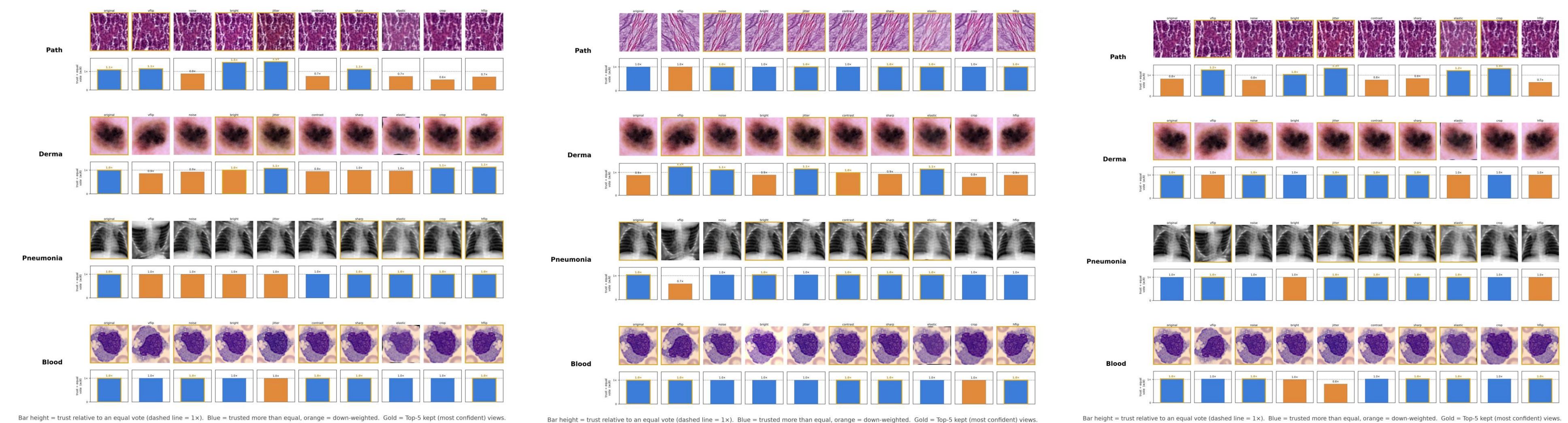
mean accuracy change (Wilcoxon, 6 datasets)

p = 0.165

Wilcoxon signed-rank not significant

≤ 2 / 6

datasets with a significant McNemar accuracy gain



Conclusions and Recommendation

Calibration, not Accuracy

Uncertainty-weighted TTA leaves accuracy unchanged (a deliberate, tested null) while improving ECE on the multiclass modalities most with TS + Entropy.

A clear recommendation

TS + Entropy consistently offers the best calibration across multiple datasets and backbones. Soft weighting is superior to hard Top-K filtering, and the inverse variance formula should be avoided

Honest, reproducible science

Results fully validated across 5 environments using locked 3-seed evaluation, automated unit tests, and identical evaluation metrics, ensuring our findings are robust and trustworthy.

Why it matters for diagnostics: Trustworthy confidence enables safe triage so confident cases handled automatically, uncertain ones flagged for a clinician at near-zero added cost with TS-only.

Future Directions

- Larger backbones & real hospital data (beyond 64×64)
- Train the model to automatically score the quality of each augmented view
- Edge / streaming deployment TS-only is the cheap path
- Add mathematical guarantees to our confidence scores for safer clinical use